



PROBLEM SOLVING WITH C

UE25CS151B

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PROBLEM SOLVING WITH C

C Programming Environment



1. Installation of gcc on different Operating systems
2. Program Development Life Cycle [PDLC]
3. First Program in C
4. Structure of C Program
5. Steps involved in execution of C Program
6. C Compiler standards
7. Errors during execution

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Installation of gcc on different Operating systems

- Windows OS – Installation of gcc using Mingw.
 - <https://www.youtube.com/watch?v=sXW2VLrQ3Bs>
- Linux OS – gcc available by default

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Program Development Life Cycle [PDLC]

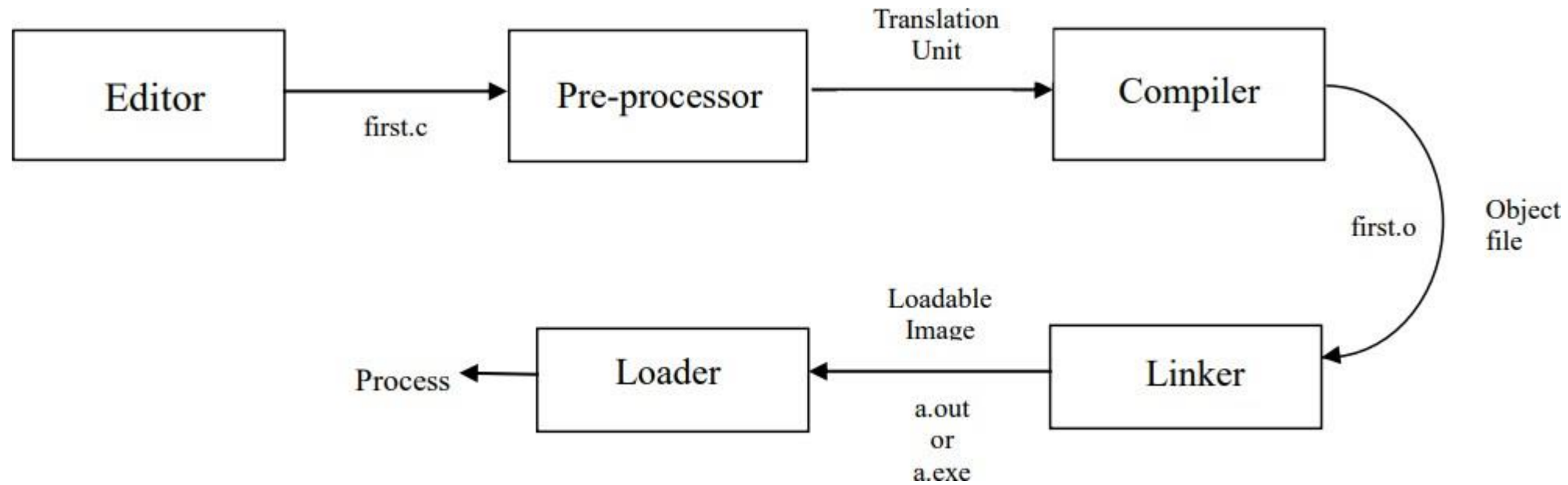


Fig 1: Phases involved in PDLC

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First Program in C

```
#include<stdio.h>
int main()
{
    printf("Hello PES\n");
    return 0;
}

//Importance of int main and return values
```

Program Structure

- Case sensitive
- Indentation is not a language requirement
- The main() is the starting point of execution
- Comments in C
 - Using // for single line comment
 - Using /* and */ for multiline comment

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Steps involved in execution of code

Way 1:

Step1: `gcc <filename> // image a.out is the output`

Step2: `./a.out` OR `a.exe`

Way 2: Creating the object files separately

Compile: `gcc -c <filename> // filename.o is the output`

Link: `gcc filename.o` OR `gcc <list of object files> -l <library>`

Execute: `./a.out` OR `a.exe`

Way 3: Renaming the object files

Step1: `gcc <filename> -o <imagename>`

Step2: `<imagename>`

C Compiler standards

Standardized by ANSI and ISO: **C89, C99 and C11**

Using c99:

```
gcc -std=c99 program.c
```

Using c11:

```
gcc -std=c11 program.c
```

Use **__STDC_VERSION__** we can get the standard C version

Errors during Execution

- Compile time Error
- Link time Error
- Run time Error
- Logical Error

// One Example on each of these



THANK YOU

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